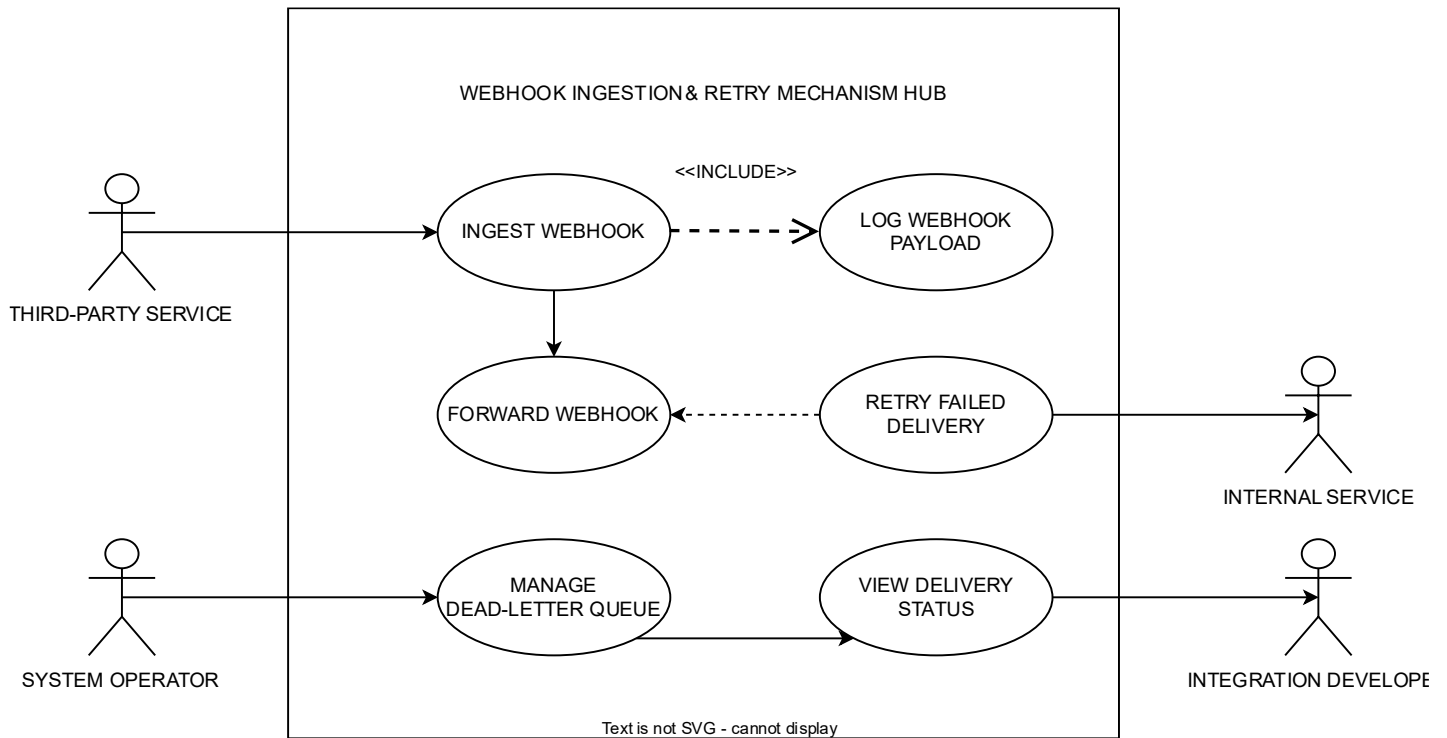


Webhook Ingestion & Retry Mechanism Hub Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ingest inbound webhooks from third-party services.	High	Pass: A valid inbound webhook is successfully received and accepted for processing. Fail: A valid webhook is not received or is rejected without processing.	Receiving third-party events is the core purpose of the system.
FR-002	Functional	The system shall log the request payload of every received webhook.	High	Pass: Each received webhook has its request payload recorded in the system logs. Fail: A received webhook has no corresponding payload log.	Logging enables tracking and troubleshooting of webhook events.
FR-003	Functional	The system shall forward received webhook payloads to the target internal service.	High	Pass: A received webhook payload is forwarded to the target endpoint and the delivery result is recorded. Fail: The payload is not forwarded to the target endpoint.	Delivering webhook events to internal services is a primary system function.
FR-004	Functional	The system shall automatically retry failed webhook deliveries using exponential backoff intervals of 1s, 2s, 4s, and 8s.	High	Pass: A failed delivery is retried automatically at the specified intervals and each retry is logged. Fail: A failed delivery is not retried according to the specified intervals.	Automatic retries improve delivery reliability during temporary failures.
FR-005	Functional	The system shall move webhook deliveries that remain unsuccessful after all retry attempts to a dead-letter queue.	High	Pass: A webhook that fails all retry attempts is persisted in the dead-letter queue. Fail: A permanently failed webhook is dropped without dead-letter persistence.	Prevents permanent loss of failed webhook events.
NFR-001	Non-Functional	The ingestion endpoint shall buffer inbound webhooks with 99.999% reliability and throughput up to 2,500 requests/second.	High	Pass: Benchmarking tests confirm the specified reliability and throughput under simulated peak load. Fail: The system fails to meet either target.	Ensures reliable handling of a high volume of webhook events.
NFR-002	Non-Functional	The system shall protect inbound webhook requests and logged payload data from unauthorized access.	High	Pass: Security testing confirms unauthorized users cannot access protected webhook data or logs. Fail: Unauthorized access to webhook requests or payload logs is possible.	Webhook payloads may contain sensitive data that must be protected.



Use-Case Flow Specification

Use Case ID: UC-03

Use Case Name: Forward Webhook

Preconditions

1. A webhook has been successfully received by the system.
2. The webhook request payload has been logged.
3. A target internal service is configured and available.

Postconditions

1. The webhook payload is successfully delivered to the internal service and the delivery result is recorded; or
2. If all retry attempts fail, the webhook delivery is moved to the Dead-Letter Queue for further management.

Main Success Scenario

1. The system selects the received webhook payload for delivery.
2. The system forwards the payload to the configured internal service.
3. The internal service receives and processes the webhook payload.
4. The internal service returns a successful response.

5. The system records the delivery as successful.
6. The use case ends successfully.

Alternate Flow — Delivery Failure

At Step 4:

1. The internal service does not return a successful response.
2. The system identifies the webhook delivery as failed.
3. The system initiates Retry Failed Delivery using exponential backoff intervals of 1s, 2s, 4s, and 8s.
4. If a retry succeeds, the system records the delivery as successful and the use case ends.
5. If all retry attempts fail, the system moves the webhook delivery to the Dead-Letter Queue for further management.